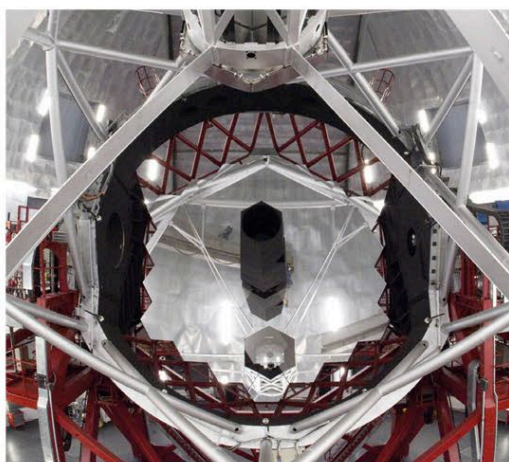


NEW OPTICAL TECHNOLOGY

In their quest for greater light grasp and sharper images, optical astronomers have utilized innovative new technology, such as mirrors made up of many separate segments. Segmented mirrors can be made much thinner, and hence lighter, than a single large mirror. The segments are usually hexagonal in shape, and each one can be individually controlled to maintain sharp focus as the telescope is moved. Mirrors larger than 26.2 ft (8 m) in diameter are now made in this way, and segmented mirrors up to 128 ft (39 m) wide are planned. Another advance has come from adaptive optics, a technique that removes the blurring effects of the atmosphere and can produce images almost as sharp as those from telescopes in space. This is done by measuring atmospheric distortion using an artificial guide star created by firing a laser beam along the telescope's line of sight. Using these measurements, a flexible secondary mirror (which collects the light from the main mirror) is then deformed to compensate for the distortion.

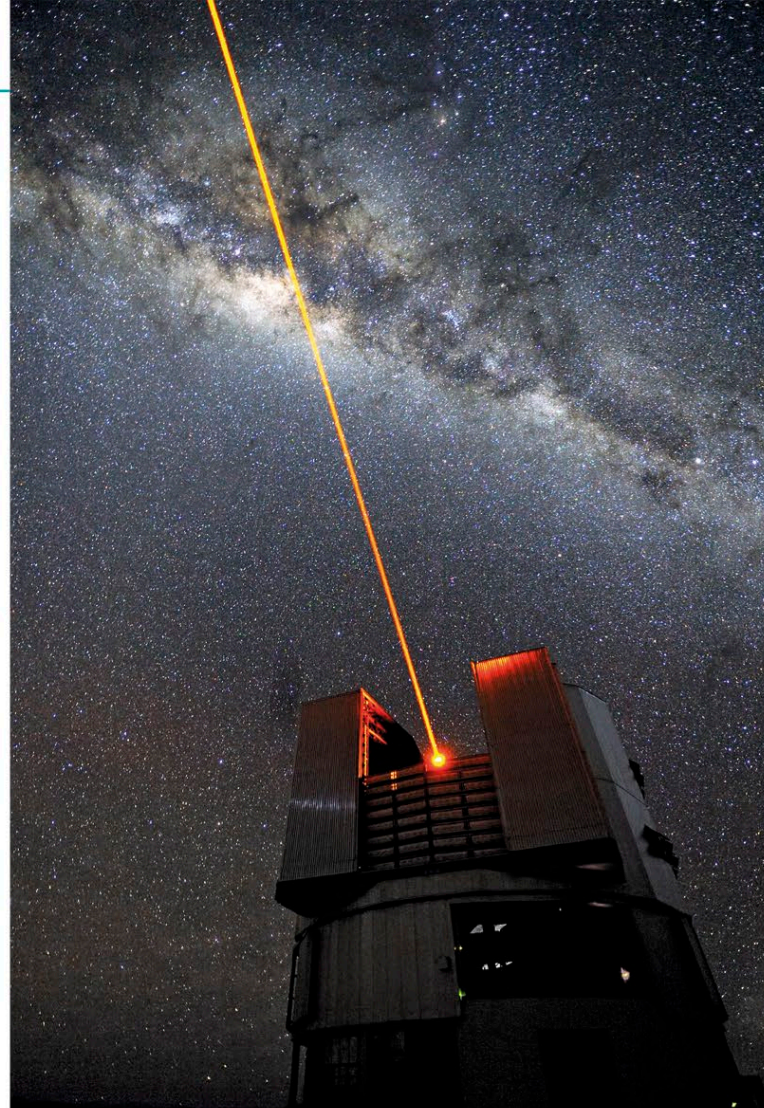


SEGMENTED TELESCOPE MIRROR

The Gran Telescopio Canarias, also known as the GranTeCan or GTC, has a mirror 34 ft (10.4 m) in diameter—the world's largest. Opened in 2009, it is located at the Roque de los Muchachos on La Palma in the Canary Islands. Its mirror (shown left) is composed of 36 hexagonal segments, each of which is 75 in (1.9 m) wide.

THE LARGE BINOCULAR TELESCOPE

A novel design for increasing light grasp and resolving power is the Large Binocular Telescope at Mount Graham, Arizona. It consists of two mirrors, each 27.6 ft (8.4 m) in diameter, side by side on the same mount. Together, the two mirrors collect as much light as a single mirror 38.7 ft (11.8 m) across.

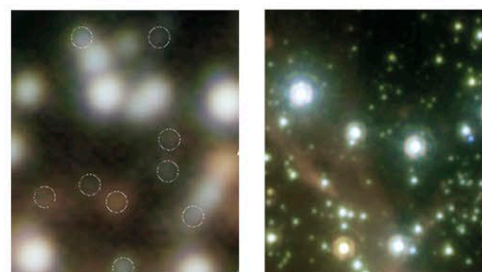


LASER GUIDE STAR

A powerful beam of orange laser light shoots skyward from one of the components of the Very Large Telescope (VLT) in Chile, creating an artificial guide star 55 miles (90 km) high. The guide star is part of the VLT's adaptive optics system, which helps correct for image distortion caused by atmospheric disturbances.

EFFECT OF ADAPTIVE OPTICS

These images of the center of the Galaxy through the Keck II telescope in Hawaii show the effect of adaptive optics. The image on the left was taken without adaptive optics; the much sharper image on the right was taken with the adaptive optics system in operation.



BEYOND VISIBLE LIGHT

Many celestial objects emit energy outside the visible light spectrum (see pp.36–37), so optical telescopes alone cannot give a complete view. The first non-visible-light telescope was a radio telescope, built in 1937. Radio waves have much longer wavelengths than visible light, so radio telescopes have to be larger to achieve the same resolution. To overcome this restriction, radio dish arrays have been built so that observations from individual dishes can be combined. An example is the Karl G. Jansky Very Large Array near Socorro, New Mexico, which consists of 27 dishes, each 82 ft (25 m) wide, arranged along three arms 13 miles (21 km) long. The largest single radio dish is 1,000 ft (305 m) in diameter at Arecibo, Puerto Rico. Most non-visible-light wavelengths other than radio are blocked by the atmosphere. However, some infrared reaches mountaintops and can be detected by certain telescopes, such as the United Kingdom Infrared Telescope in Hawaii. It is also possible to detect cosmic gamma rays at the Earth's surface. The MAGIC telescope at La Palma in the Canary Islands achieves this by detecting the faint light emitted by particle showers generated by gamma rays.

GREEN BANK RADIO TELESCOPE

The world's largest fully steerable radio telescope, at the National Radio Astronomy Observatory at Green Bank, West Virginia, has an elliptical dish 360 x 328 ft (110 x 100 m) across. The dish consists of over 2,000 panels, each of which can be adjusted separately to maintain the shape of the dish as the telescope moves. The secondary reflector (which reflects radio waves from the main dish to the radio detector) is on an arm to avoid obstructing the main dish.

